CONTRACT WOLF

Security Assessment

Mineral Project Coin

Verified on October 15, 2026

SUMMARY

Project	CHAIN	METHODOLOGY
Mineral Project Coin	opBNB	Manual & Automatic Analysis

FILES	DELIVERY	TYPE
Live Contract	October 15, 2026	Standard Audit



11	0	0	0	0	11	11
Total Findings	Critical	Major	Medium	Minor	Informational	Acknowledged

🟥 0 Critical	An exposure that can affect the contract functions in several events that can risk and disrupt the contract
🟧 0 Major	An opening & exposure to manipulate the contract in an unwanted manner
🟨 0 Medium	An opening that could affect the outcome in executing the contract in a specific situation
⬛ 0 Minor	An opening but doesn't have an impact on the functionality of the contract
🟦 11 Informational	An opening that consists information but will not risk or affect the contract
🟩 11 Acknowledged	ContractWolf's findings has been acknowledged & resolved by the project

STATUS ✔️**AUDIT PASSED**

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DISCLAIMER | Mineral Project Coin

ContractWolf audits and reports should not be considered as a form of project's "Advertisement" and does not cover any interaction and assessment from "Project Contract" to "External Contracts" such as PancakeSwap, UniSwap, SushiSwap or similar.

ContractWolf does not provide any warranty on its released report and should not be used as a decision to invest into audited projects.

ContractWolf provides a transparent report to all its "Clients" and to its "Clients Participants" and will not claim any guarantee of bug-free code within its **SMART CONTRACT**.

ContractWolf's presence is to analyze, audit and assess the Client's Smart Contract to find any underlying risk and to eliminate any logic and flow errors within its code.

Each company or project should be liable to its security flaws and functionalities.

SCOPE OF WORK | Mineral Project Coin

Mineral Project Coin team has agreed and provided us with the files that need to be tested (*Github, BSCscan, Etherscan, Local files etc*). The scope of audit is the main contract.

The goal of this engagement is to identify if there is a possibility of security flaws in the implementation of smart contract and its systems.

ContractWolf will be focusing on contract issues and functionalities along with the project claims from smart contract to their website, whitepaper, repository which has been provided by **Mineral Project Coin**.

AUDITING APPROACH | Mineral Project Coin

Every line of code along with its functionalities will undergo manual review to check for security issues, quality of logic and contract scope of inheritance. The manual review will be done by our team that will document any issues that they discovered.

METHODOLOGY

The auditing process follows a routine series of steps :

1. Code review that includes the following :
   - Review of the specifications, sources and instructions provided to ContractWolf to make sure we understand the size, scope and functionality of the smart contract.
   - Manual review of code. Our team will have a process of reading the code line-by-line with the intention of identifying potential vulnerabilities, underlying and hidden security flaws.

2. Testing and automated analysis that includes :
   - Testing the smart contract function with common test cases and scenarios to ensure that it returns the expected results.

3. Best practices and ethical review. The team will review the contract with the aim to improve efficiency, effectiveness, clarifications, maintainability, security and control within the smart contract.

4. Recommendations to help the project take steps to eliminate or minimize threats and secure the smart contract.

TOKEN DETAILS | Mineral Project Coin



Mining RWA Infrastructure, Mineral Project Coin (MPC) is a blockchain-based Real World Asset (RWA) infrastructure designed to standardize the tokenization of mining assets.

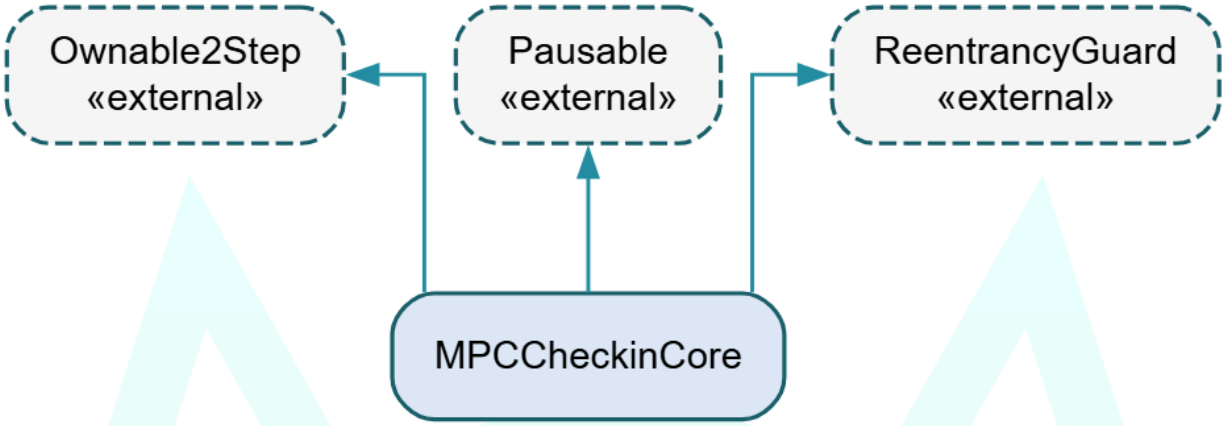
Token Name	Symbol	Decimal	Total Supply	Chain
MPCCheckinCore	MPC	N/A	N/A	opBNB

SOURCE

Source	*Live Contract (0x843e4ff9242c3e81b676434b2876fb090d4a5783)*

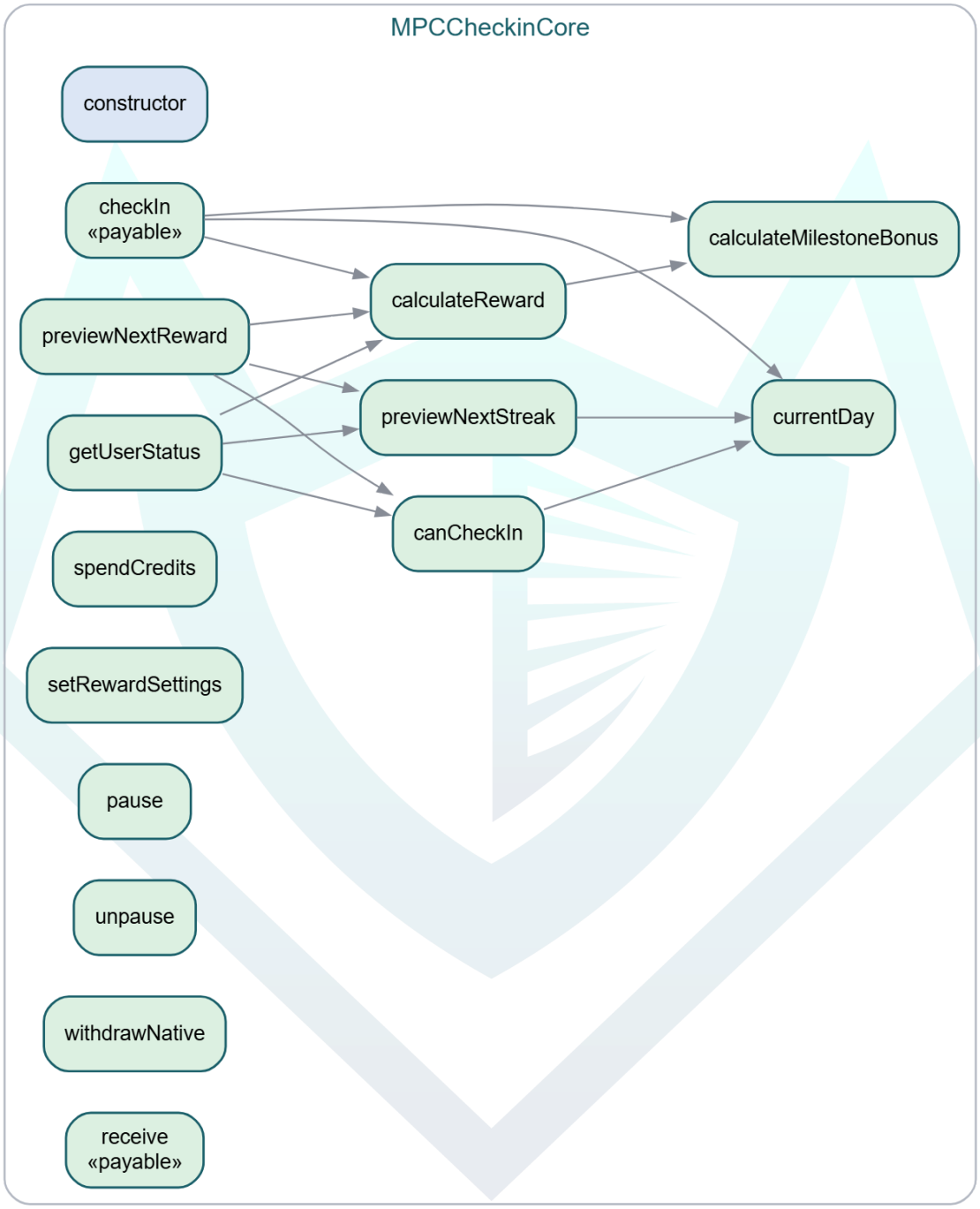
INHERITANCE GRAPH | Mineral Project Coin

Inheritance Graph of Contract Functions



CALL GRAPH | Mineral Project Coin

Call Graph of Contract Functions



FINDINGS | Mineral Project Coin



11	0	0	0	0	11	11
Total Findings	Critical	Major	Medium	Minor	Informational	Acknowledged

This report has been prepared to state the issues and vulnerabilities for Mineral Project Coin through this audit. The goal of this report findings is to identify specifically and fix any underlying issues and errors

ID	Title	File & Line #	Severity	Status
SYBIL-FARM-01	Credit issuance has no Sybil resistance	MPCCheckinCore.sol, L: 164	Informational	Acknowledged
CW-002	Reward parameters are owner-settable with no bounds	MPCCheckinCore.sol, L: 519	Informational	Acknowledged
CW-002 (2)	renounceOwnership() freezes admin functions and strands BNB	MPCCheckinCore.sol, L: 147	Informational	Acknowledged
SWC-114	Reward settings can be front-run against a pending check-in	MPCCheckinCore.sol, L: 519	Informational	Acknowledged
CREDIT-REDEEM-01	spendCredits() burns credits with no redemption context and no refund path	MPCCheckinCore.sol, L: 483	Informational	Acknowledged
SWC-116	Day boundary and streak continuity rest on block.timestamp	MPCCheckinCore.sol, L: 352	Informational	Acknowledged
PAYABLE-CHECKIN-01	checkIn() is payable with no functional use for the value sent	MPCCheckinCore.sol, L: 166	Informational	Acknowledged
EVENT-DOUBLECOUNT-01	StreakBonusAwarded re-reports credits already counted in CheckedIn	MPCCheckinCore.sol, L: 248	Informational	Acknowledged

CW-006	calculateMilestoneBonus() is computed twice on every check-in	MPCCheckinCore.sol, L: 248	Informational	Acknowledged
CW-010	Global and per-user counters share near-identical names	MPCCheckinCore.sol, L: 63	Informational	Acknowledged
PAUSE-PREVIEW-01	View helpers report check-in as available while the contract is paused	MPCCheckinCore.sol, L: 358	Informational	Acknowledged

SWC ATTACKS | Mineral Project Coin

Smart Contract Weakness Classification and Test Cases

ID	Description	Status
SWC-100	Function Default Visibility	Passed
SWC-101	Integer Overflow and Underflow	Passed
SWC-102	Outdated Compiler Version	Passed
SWC-103	Floating Pragma	Passed
SWC-104	Unchecked Call Return Value	Passed
SWC-105	Unprotected Ether Withdrawal	Passed
SWC-106	Unprotected SELF DESTRUCT Instruction	Passed
SWC-107	Reentrancy	Passed
SWC-108	State Variable Default Visibility	Passed
SWC-109	Uninitialized Storage Pointer	Passed
SWC-110	Assert Violation	Passed
SWC-111	Use of Deprecated Solidity Functions	Passed
SWC-112	Delegate call to Untrusted Callee	Passed
SWC-113	DoS with Failed Call	Passed
SWC-114	Transaction Order Dependence	Passed
SWC-115	Authorization through tx.origin	Passed
SWC-116	Block values as a proxy for time	Passed
SWC-117	Signature Malleability	Passed
SWC-118	Incorrect Constructor Name	Passed
SWC-119	Shadowing State Variables	Passed
SWC-120	Weak Sources of Randomness from Chain Attributes	Passed
SWC-121	Missing Protection against Signature Replay Attacks	Passed
SWC-122	Lack of Proper Signature Verification	Passed

ID	Description	Status
SWC-123	Requirement Violation	Passed
SWC-124	Write to Arbitrary Storage Location	Passed
SWC-125	Incorrect Inheritance Order	Passed
SWC-126	Insufficient Gas Griefing	Passed
SWC-127	Arbitrary Jump with Function Type Variable	Passed
SWC-128	DoS With Block Gas Limit	Passed
SWC-129	Typographical Error	Passed
SWC-130	Right-To-Left-Override control character(U+202E)	Passed
SWC-131	Presence of unused variables	Passed
SWC-132	Unexpected Ether balance	Passed
SWC-133	Hash Collisions With Multiple Variable Arguments	Passed
SWC-134	Message call with hardcoded gas amount	Passed
SWC-135	Code With No Effects	Passed
SWC-136	Unencrypted Private Data On-Chain	Passed

CW ASSESSMENT | Mineral Project Coin

ContractWolf Vulnerability and Security Tests

ID	Name	Description	Status
CW-001	Multiple Version	Presence of multiple compiler version across all contracts	✔
CW-002	Incorrect Access Control	Additional checks for critical logic and flow	✔
CW-003	Payable Contract	A function to withdraw ether should exist otherwise the ether will be trapped	✔
CW-004	Custom Modifier	major recheck for custom modifier logic	✔
CW-005	Divide Before Multiply	Performing multiplication before division is generally better to avoid loss of precision	✔
CW-006	Multiple Calls	Functions with multiple internal calls	✔
CW-007	Deprecated Keywords	Use of deprecated functions/operators such as block.blockhash() for blockhash(), msg.gas for gasleft(), throw for revert(), sha3() for keccak256(), callcode() for delegatecall(), suicide() for selfdestruct(), constant for view or var for actual type name should be avoided to prevent unintended errors with newer compiler versions	✔
CW-008	Unused Contract	Presence of an unused, unimported or uncalled contract	✔
CW-009	Assembly Usage	Use of EVM assembly is error-prone and should be avoided or double-checked for correctness	✔
CW-010	Similar Variable Names	Variables with similar names could be confused for each other and therefore should be avoided	✔
CW-011	Commented Code	Removal of commented/unused code lines	✔
CW-012	SafeMath Override	SafeMath is no longer needed starting with Solidity v0.8+. The compiler now has built-in overflow checking.	✔

FIXES & RECOMMENDATION

SYBIL-FARM-01 | Credit issuance has no Sybil resistance

`checkIn()` is permissionless and costs only gas. Its single gate, `hasCheckedInOnDay[msg.sender][today]`, limits each address to one check-in per day but puts no cost on creating addresses. One operator can fund thousands of wallets and collect `baseReward` from each of them daily, at a fraction of a cent of gas per call on opBNB. `totalUsers` and `totalCreditsIssued` are therefore not trustworthy, and any off-chain value attached to credits is farmable at scale.

```
function checkIn()
    external
    payable
    whenNotPaused
    nonReentrant
    returns (uint256 streak, uint256 credits)
{
    uint256 today = currentDay();

    // The only gate: one per address per day. Addresses are free.
    if (hasCheckedInOnDay[msg.sender][today]) {
        revert AlreadyCheckedInToday();
    }
    ...
}
```

Recommendation

Tie issuance to something an attacker cannot mint for free. The lightest working option is a per-wallet signature from a backend key that has already run the project's anti-abuse checks, so an address alone is not enough. A non-transferable badge or a small `msg.value` are alternatives. Until then, treat `totalUsers` as an upper bound, not a user count.

```
address public signerAuthority;
mapping(address => uint256) public checkInNonce;

error InvalidSignature();

function checkIn(bytes calldata signature)
    external
    whenNotPaused
    nonReentrant
    returns (uint256 streak, uint256 credits)
{
    uint256 today = currentDay();

    if (hasCheckedInOnDay[msg.sender][today]) {
        revert AlreadyCheckedInToday();
    }

    bytes32 digest = MessageHashUtils.toEthSignedMessageHash(
        keccak256(
            abi.encode(
                block.chainid,
                address(this),
                msg.sender,
                today,
                checkInNonce[msg.sender]++
            )
        )
    );

    if (ECDSA.recover(digest, signature) != signerAuthority) {
        revert InvalidSignature();
    }

    // ... unchanged streak and credit accounting ...
}
```

CW-002 | Reward parameters are owner-settable with no bounds

`setRewardSettings()` only checks that the two thresholds are non-zero. The other four values are stored unchecked, with no cap and no timelock. A very large `streakRewardIncrement` makes `calculateReward()` overflow and revert under Solidity `0.8.x`, so `checkIn()` fails for every wallet with a streak of 2 or more until the owner undoes it. A threshold of 1 pays `weeklyBonus` on every check-in.

```
function setRewardSettings(
    uint256 newBaseReward,
    uint256 newStreakRewardIncrement,
    uint256 newWeeklyBonusThreshold,
    uint256 newMonthlyBonusThreshold,
    uint256 newWeeklyBonus,
    uint256 newMonthlyBonus
)
    external
    onlyOwner
{
    if (newWeeklyBonusThreshold == 0 || newMonthlyBonusThreshold == 0) {
        revert InvalidRewardConfiguration();
    }

    baseReward = newBaseReward;                         // unbounded
    streakRewardIncrement = newStreakRewardIncrement;   // unbounded: bricks checkIn()
    weeklyBonusThreshold = newWeeklyBonusThreshold;     // may be 1
    monthlyBonusThreshold = newMonthlyBonusThreshold;   // may be 1
    weeklyBonus = newWeeklyBonus;                       // unbounded
    monthlyBonus = newMonthlyBonus;                     // unbounded
    ...
}
```

Recommendation

Add hard ceilings as constants so no owner key can push the reward maths into a reverting range or turn the weekly bonus into a daily payout. Require both thresholds to be at least 2 and cap each credit value at a figure the project commits to publicly.

```
uint256 public constant MAX_BASE_REWARD = 1_000;
uint256 public constant MAX_STREAK_INCREMENT = 100;
uint256 public constant MAX_BONUS = 10_000;
uint256 public constant MIN_THRESHOLD = 2;

function setRewardSettings(
    uint256 newBaseReward,
    uint256 newStreakRewardIncrement,
    uint256 newWeeklyBonusThreshold,
    uint256 newMonthlyBonusThreshold,
    uint256 newWeeklyBonus,
    uint256 newMonthlyBonus
)
    external
    onlyOwner
{
    if (
        newWeeklyBonusThreshold < MIN_THRESHOLD ||
        newMonthlyBonusThreshold < MIN_THRESHOLD ||
        newBaseReward > MAX_BASE_REWARD ||
        newStreakRewardIncrement > MAX_STREAK_INCREMENT ||
        newWeeklyBonus > MAX_BONUS ||
        newMonthlyBonus > MAX_BONUS
    ) {
        revert InvalidRewardConfiguration();
    }

    // ... assignments unchanged ...
}
```

CW-002 (2) | renounceOwnership() freezes admin functions and strands BNB

`Ownable2Step` makes `transferOwnership()` a two-step handshake but leaves `renounceOwnership()` from `Ownable` as a single unconfirmed call that sets `owner()` to `address(0)`. After that, `pause()`, `unpause()`, `setRewardSettings()` and `withdrawNative()` can never run again: reward values freeze, the emergency stop is lost, and any BNB the contract holds is stranded permanently.

```
contract MPCCheckinCore is
    Ownable2Step,     // hardens transferOwnership, but NOT renounceOwnership
    Pausable,
    ReentrancyGuard
{
    constructor() Ownable(msg.sender) {}

    // renounceOwnership() is inherited from Ownable and is a single,
    // unconfirmed call that sets owner() to address(0) forever.
}
```

Recommendation

Override `renounceOwnership()` to revert unless the project truly intends an ownerless contract. If it does, require a zero BNB balance and an explicit confirmation argument first, so it cannot happen through a mistyped call.

```
error RenounceDisabled();

/// @notice Ownership cannot be renounced: doing so would permanently
///         freeze pause control, reward settings and BNB withdrawal.
function renounceOwnership() public view override onlyOwner {
    revert RenounceDisabled();
}
```

SWC-114 | Reward settings can be front-run against a pending check-in

Credits are read from mutable storage inside `calculateReward()` at execution time, not fixed when the user signs. `previewNextReward()` may quote 72, then the owner's `setRewardSettings()` lands first in the same block and drops `baseReward` and `weeklyBonus` to `0`. The user's transaction still succeeds, burns their once-per-day slot, and mints almost nothing. `checkIn()` accepts no minimum, so the user cannot protect themselves.

```
// Reward is resolved against live storage, at execution time:
uint256 earnedCredits = calculateReward(newStreak);

// ...and setRewardSettings() may be mined immediately before it:
function setRewardSettings(...) external onlyOwner {
    baseReward = newBaseReward;
    weeklyBonus = newWeeklyBonus;
    ...
}
```

Recommendation

Let the caller pass the minimum credits they will accept, so a reward cut mined ahead of them reverts instead of silently consuming their daily slot. Pair this with a timelock on `setRewardSettings()` so changes are visible before they bind.

```
error RewardBelowMinimum(uint256 actual, uint256 minimum);

function checkIn(uint256 minCredits)
    external
    whenNotPaused
    nonReentrant
    returns (uint256 streak, uint256 credits)
{
    // ... streak calculation unchanged ...

    uint256 earnedCredits = calculateReward(newStreak);

    if (earnedCredits < minCredits) {
        revert RewardBelowMinimum(earnedCredits, minCredits);
    }

    // ... accounting unchanged ...
}
```

CREDIT-REDEEM-01 | spendCredits() burns credits with no redemption context and no refund path

`spendCredits()` takes only an `amount`. It debits `availableCredits` and emits `CreditsSpent` with no item, order id or recipient, so a backend cannot tell what was redeemed or separate two identical spends in one block. The burn is also final: nothing but `checkIn()` ever increases `availableCredits`, so if fulfilment fails after the transaction confirms, neither the user nor the owner can restore the balance.

```
function spendCredits(uint256 amount)
    external
    whenNotPaused
    nonReentrant
{
    if (amount == 0) {
        revert ZeroAmount();
    }

    UserInfo storage user = users[msg.sender];

    if (user.availableCredits < amount) {
        revert InsufficientCredits();
    }

    user.availableCredits -= amount;

    // No reference to what was bought; no way to undo.
    emit CreditsSpent(msg.sender, amount, user.availableCredits);
}
```

Recommendation

Carry an opaque `reference` through the call and index it on the event, so the backend can match each burn to its order and reject replays. Add a bounded, owner-only `refundCredits()` with its own event for the failed-fulfilment case.

```
event CreditsSpent(
    address indexed user,
    bytes32 indexed reference,
    uint256 amount,
    uint256 remainingCredits
);

event CreditsRefunded(
    address indexed user,
    bytes32 indexed reference,
    uint256 amount
);

function spendCredits(uint256 amount, bytes32 reference)
    external
    whenNotPaused
    nonReentrant
{
    if (amount == 0) {
        revert ZeroAmount();
    }

    UserInfo storage user = users[msg.sender];

    if (user.availableCredits < amount) {
        revert InsufficientCredits();
    }

    user.availableCredits -= amount;

    emit CreditsSpent(msg.sender, reference, amount, user.availableCredits);
}

function refundCredits(address account, uint256 amount, bytes32 reference)
    external
    onlyOwner
{
    if (amount == 0) {
        revert ZeroAmount();
    }

    users[account].availableCredits += amount;

    emit CreditsRefunded(account, reference, amount);
}
```

SWC-116 | Day boundary and streak continuity rest on block.timestamp

`currentDay()` returns `block.timestamp / 1 days`, so eligibility, the `hasCheckedInOnDay` key and the `lastCheckInDay + 1 == today` streak test all depend on a timestamp the opBNB sequencer chooses. A block near UTC midnight can land on either side of the boundary. The direct effect is small, but the streak test is exact, so one shifted boundary resets a long streak to 1 and forfeits the weekly and monthly bonuses.

```
function currentDay() public view returns (uint256) {
    return block.timestamp / 1 days;
}

// Exact equality: one shifted boundary resets the streak to 1.
} else if (user.lastCheckInDay + 1 == today) {
    newStreak = user.currentStreak + 1;
} else {
    newStreak = 1;
}
```

Recommendation

Keep the timestamp-derived day, since there is no better on-chain clock, but stop making streak continuity depend on it exactly. Treating a one-day gap as continuing the streak absorbs a boundary shift, while `hasCheckedInOnDay` still caps issuance at one per day.

```
uint256 constant STREAK_GRACE_DAYS = 1;

// Continue the streak when the gap is within the grace window,
// so a shifted day boundary cannot destroy a long streak.
uint256 gap = today - user.lastCheckInDay;

if (user.totalCheckIns == 0) {
    newStreak = 1;
} else if (gap <= 1 + STREAK_GRACE_DAYS) {
    newStreak = user.currentStreak + 1;
} else {
    newStreak = 1;
}
```

PAYABLE-CHECKIN-01 | checkIn() is payable with no functional use for the value sent

`checkIn()` is `payable`, yet nothing prices, refunds or credits `msg.value`. It is absorbed into the contract balance and only logged by `NativeReceived`. The contract's own comments say BNB is not required, so any value sent here is a mistake, typically from a mis-built front end. The call succeeds silently, and only the owner can recover the funds through `withdrawNative()`.

```
function checkIn()
    external
    payable          // accepts BNB the contract has no use for
    whenNotPaused
    nonReentrant
    returns (uint256 streak, uint256 credits)
{
    ...
    // Value is logged and kept; never priced, never refunded.
    if (msg.value > 0) {
        emit NativeReceived(msg.sender, msg.value);
    }
}
```

Recommendation

Drop `payable` and the `msg.value` block. A value-bearing call then reverts and the user keeps their BNB, which is the right outcome for a function documented as free. Deliberate donations still have `receive()`.

```
function checkIn()
    external
    whenNotPaused
    nonReentrant
    returns (uint256 streak, uint256 credits)
{
    // No `payable`: a call carrying value now reverts and the
    // sender keeps their BNB. Donations still go through receive().
    ...
}
```

EVENT-DOUBLECOUNT-01 | StreakBonusAwarded re-reports credits already counted in CheckedIn

`calculateReward()` already includes `calculateMilestoneBonus()` in its result, so the `credits` value in `CheckedIn` is the full amount, bonus included. `checkIn()` then recomputes the same bonus and emits it again in `StreakBonusAwarded`. On-chain totals are correct, but nothing in the two events says the bonus is a subset of `credits` rather than an addition, so an indexer that sums both overstates issuance on every 7th and 30th day of a streak.

```
// calculateReward() already includes the milestone bonus:
uint256 reward = baseReward + ((streak - 1) * streakRewardIncrement);
reward += calculateMilestoneBonus(streak);

// ...so `earnedCredits` here is the full amount, bonus included:
emit CheckedIn(msg.sender, today, newStreak, earnedCredits);

// ...and this emits the same bonus a second time:
uint256 milestoneBonus = calculateMilestoneBonus(newStreak);

if (milestoneBonus > 0) {
    emit StreakBonusAwarded(msg.sender, today, newStreak, milestoneBonus);
}
```

Recommendation

Make the two figures disjoint so consumers can add them safely: emit base and bonus as separate fields on `CheckedIn`, and keep `StreakBonusAwarded` as a milestone marker rather than a second ledger entry.

```
event CheckedIn(
    address indexed user,
    uint256 indexed dayIndex,
    uint256 streak,
    uint256 baseCredits,   // excludes the milestone bonus
    uint256 bonusCredits   // the milestone bonus, counted once
);

(uint256 baseCredits, uint256 bonusCredits) = _splitReward(newStreak);
uint256 earnedCredits = baseCredits + bonusCredits;

emit CheckedIn(msg.sender, today, newStreak, baseCredits, bonusCredits);

if (bonusCredits > 0) {
    emit StreakBonusAwarded(msg.sender, today, newStreak, bonusCredits);
}
```

CW-006 | calculateMilestoneBonus() is computed twice on every check-in

`checkIn()` calls `calculateReward()`, which calls `calculateMilestoneBonus(streak)`, then calls `calculateMilestoneBonus(newStreak)` again with the same argument only to decide whether to emit `StreakBonusAwarded`. Both reads touch four storage slots, so every check-in pays four extra warm `SLOAD` operations on the contract's only hot path.

```
// First computation, inside calculateReward():
uint256 earnedCredits = calculateReward(newStreak);

// ... 40 lines later, the identical call again:
uint256 milestoneBonus = calculateMilestoneBonus(newStreak);

if (milestoneBonus > 0) {
    emit StreakBonusAwarded(msg.sender, today, newStreak, milestoneBonus);
}
```

Recommendation

Compute the split once and reuse both halves. An internal helper returning base and bonus separately removes the duplicate reads and also supplies the values the `EVENT-DOUBLECOUNT-01` fix needs.

```
function _splitReward(uint256 streak)
    internal
    view
    returns (uint256 baseCredits, uint256 bonusCredits)
{
    if (streak == 0) {
        return (0, 0);
    }

    baseCredits = baseReward + ((streak - 1) * streakRewardIncrement);
    bonusCredits = calculateMilestoneBonus(streak);
}

// calculateReward() stays as the public view API:
function calculateReward(uint256 streak) public view returns (uint256) {
    (uint256 baseCredits, uint256 bonusCredits) = _splitReward(streak);
    return baseCredits + bonusCredits;
}
```

CW-010 | Global and per-user counters share near-identical names

`totalCheckIns` means two different things: a global state variable and a per-user field in `UserInfo`, so `getUserStatus()` has to rename its return to `userTotalCheckIns`. The per-user `totalCredits` sits next to the global `totalCreditsIssued`. Inside `checkIn()`, `user.totalCheckIns += 1;` and `totalCheckIns += 1;` differ only by receiver. Writing `totalCheckIns += 1` for both compiles cleanly, double-counts the global tally, and breaks the `user.totalCheckIns == 0` first-time check.

```
uint256 public totalUsers;
uint256 public totalCheckIns;         // global
uint256 public totalCreditsIssued;    // global

struct UserInfo {
    ...
    uint256 totalCheckIns;            // per-user, same name
    uint256 totalCredits;             // near-twin of totalCreditsIssued
    uint256 availableCredits;
}

// Twelve lines apart inside checkIn(), differing only by receiver:
user.totalCheckIns += 1;
totalCheckIns += 1;
```

Recommendation

Prefix the globals so no identifier means two things in one contract. The struct fields are already namespaced by `user.`, so renaming the globals is the smaller change.

```
uint256 public globalTotalUsers;
uint256 public globalTotalCheckIns;
uint256 public globalCreditsIssued;

struct UserInfo {
    ...
    uint256 totalCheckIns;      // now unambiguous
    uint256 lifetimeCredits;    // clearly distinct from the global
    uint256 availableCredits;
}
```

PAUSE-PREVIEW-01 | View helpers report check-in as available while the contract is paused

`canCheckIn()` returns `true` and `previewNextReward()` returns a non-zero figure while the contract is paused, because neither reads `paused()`. `checkIn()` carries `whenNotPaused` and reverts. A front end driving its button from `canCheckIn()` sends users into a transaction that always fails, and `getUserStatus()` passes the same wrong `canCheckInToday` into any API built on it, precisely during an incident.

```
function canCheckIn(address account) public view returns (bool) {
    // Ignores paused(); checkIn() carries whenNotPaused and will revert.
    return !hasCheckedInOnDay[account][currentDay()];
}
```

Recommendation

Fold the pause state into `canCheckIn()` so it answers whether a check-in would succeed right now. `previewNextReward()` and `getUserStatus()` already route through it and become correct with no further change.

```
function canCheckIn(address account) public view returns (bool) {
    if (paused()) {
        return false;
    }

    return !hasCheckedInOnDay[account][currentDay()];
}
```

AUDIT COMMENTS | Mineral Project Coin

Smart Contract audit comment for a non-technical perspective

🟢 Checking in is free: the contract never charges the user, only network gas.

🟢 Credits are not a token. They cannot be transferred, sold or traded/

🟢 The owner cannot take, move or reduce any user's credits. No function touches another wallet's balance.

🟢 There is no blacklist. The owner cannot block an individual wallet from checking in.

🟢 Ownership transfer is two-step: a new owner must accept before it takes effect, so the contract cannot be handed to a mistyped address.



CONTRACTWOLF

Blockchain Security - Smart Contract Audits